The game I will try to implement is the snake game. It will use the 16 by 16 led board, where the snake will be represented with the green led, and the player will control the snake where up is KEY3, down is KEY2, left is KEY1, and right is KEY0. The goal for the player is to guide the snake into a randomly generated apple on the board, represented by the red led. Every time the snake reaches the apple, it would gain 1 size in length. The game is over if the player hits the edge of the board or if the player runs into itself. The player wins if the snake reaches the max length of 256.

The idea to implement this is to have a main module to keep track of both the apple and snake location in a 2d array, as well as 4 registers that could keep track of the location of the head and tail of the snake. This way, every tick that the snake moves, we could update the 2d array representation of the snake location by turning off the led at the tail of the snake, where its location is already stored in the registers. There will be a clock divider to slow down the clock, to around 50khz, which would provide the clock for the led display driver and an LFSR circuit that would generate the location of the apple. The clock would then be slowed down even further to 2hz to run the game. The LFSR would generate the location of the apple once the snake eats an apple, and if the location of the generated position interferess with the position of the snake, it would generate again. There would also be a user-input module that would handle metastability of the user input as well as making sure the input only stays on for 1 tick.